

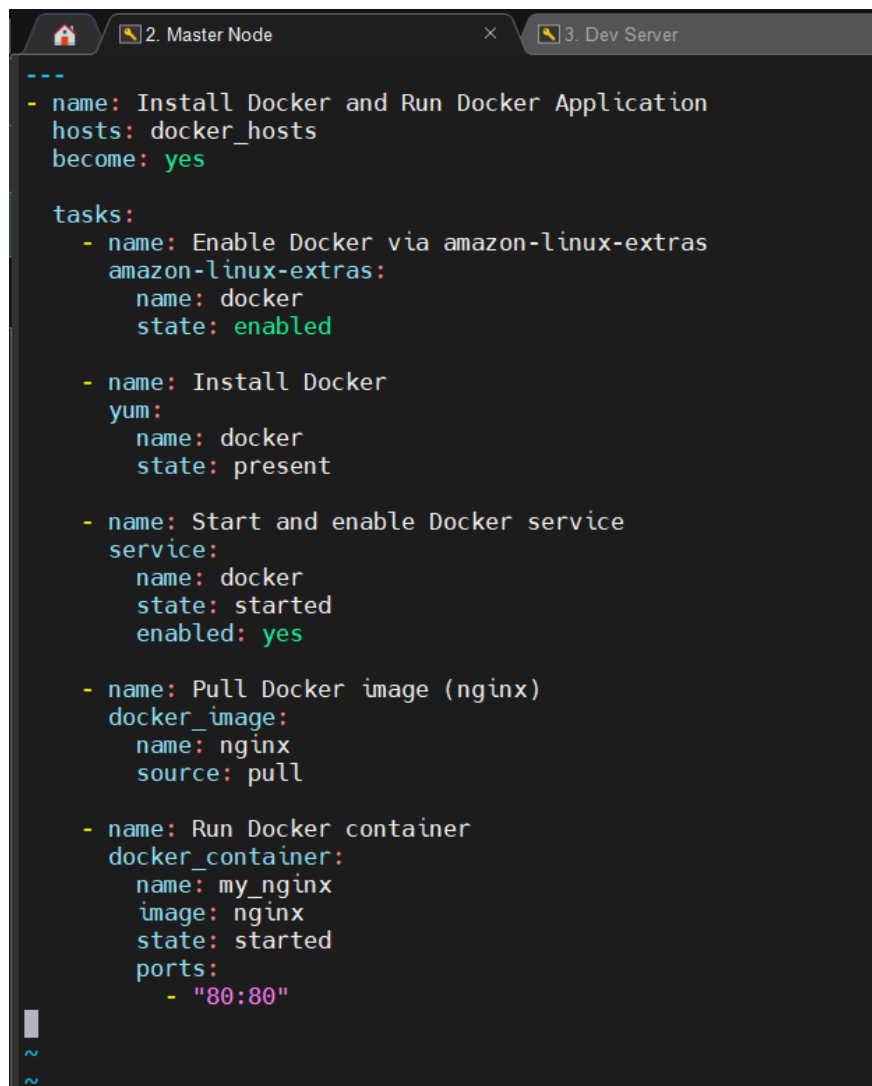
02 Assignment - Configuration

Automation using Ansible

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1. L2 - Create and Execute Ansible Playbook to Install Docker and Run the Docker Application Image created in Docker Module

- Create an Ansible playbook to:
 - install docker
 - start and enable the docker service
 - Pull the nginx Docker image
 - Run the nginx container mapping port 80

A screenshot of a code editor showing an Ansible playbook. The editor has two tabs: '2. Master Node' and '3. Dev Server'. The code is written in a dark theme with syntax highlighting. The playbook defines a play with the name 'Install Docker and Run Docker Application', targeting 'docker_hosts' and becoming the user 'yes'. It contains five tasks: enabling Docker via amazon-linux-extras, installing Docker via yum, starting and enabling the Docker service, pulling the nginx Docker image, and running a Docker container named 'my_nginx' with the nginx image, started state, and port 80 mapped to port 80.

```
---
- name: Install Docker and Run Docker Application
  hosts: docker_hosts
  become: yes

  tasks:
    - name: Enable Docker via amazon-linux-extras
      amazon-linux-extras:
        name: docker
        state: enabled

    - name: Install Docker
      yum:
        name: docker
        state: present

    - name: Start and enable Docker service
      service:
        name: docker
        state: started
        enabled: yes

    - name: Pull Docker image (nginx)
      docker_image:
        name: nginx
        source: pull

    - name: Run Docker container
      docker_container:
        name: my_nginx
        image: nginx
        state: started
        ports:
          - "80:80"
```

- Execute the playbook

```
[root@master ~]# ansible-playbook docker.yml
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details

PLAY [Install Docker and Run Docker Application] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.5.252 is using the discovered Python interpreter at
/usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for
more information.
ok: [172.31.5.252]
[WARNING]: Platform linux on host 172.31.11.110 is using the discovered Python interpreter at
/usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for
more information.
ok: [172.31.11.110]

TASK [Install Docker] *****
changed: [172.31.5.252]
changed: [172.31.11.110]

TASK [Start and enable Docker service] *****
changed: [172.31.11.110]
changed: [172.31.5.252]

TASK [Pull Docker image (nginx)] *****
changed: [172.31.11.110]
changed: [172.31.5.252]

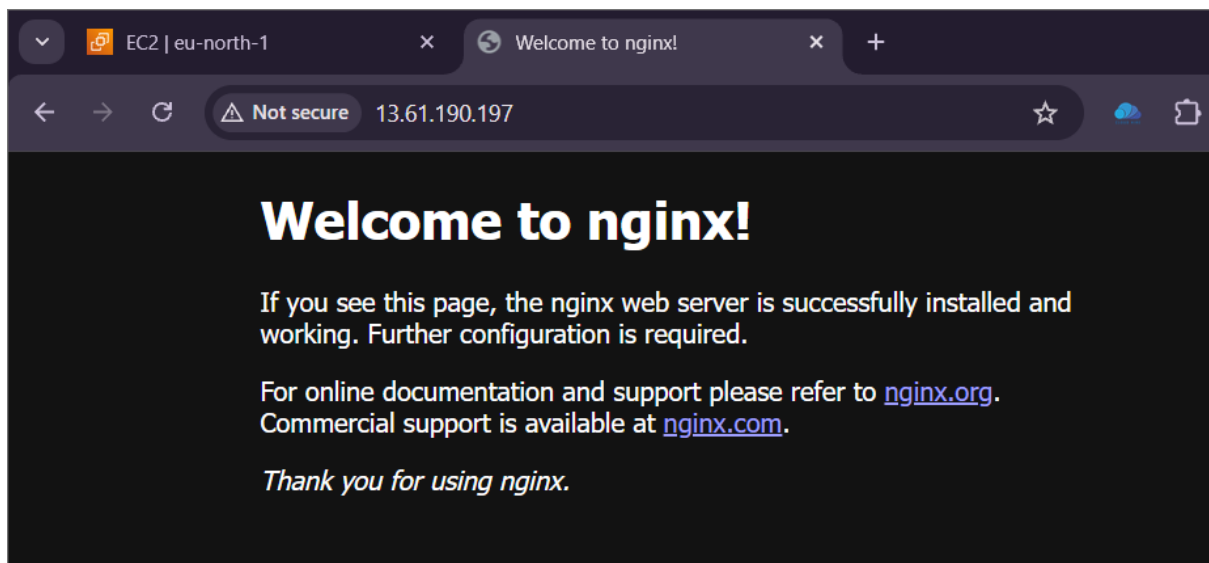
TASK [Run Docker container] *****
changed: [172.31.5.252]
changed: [172.31.11.110]

PLAY RECAP *****
172.31.11.110      : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0
172.31.5.252      : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0
[root@master ~]#
```

- Verify Docker installation and container status on Worker Node

```
[root@Dev-Server ~]# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                    NAMES
bee3f838ba4d   nginx    "/docker-entrypoint..." 2 minutes ago Up 2 minutes   0.0.0.0:80->80/tcp      my_nginx
```

- Access the nginx welcome page in a browser using the public IPs of the Worker Nodes



```
[root@master ~]# curl http://172.31.11.110
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[root@master ~]#
```

- This task demonstrated the automation of Docker installation and application deployment using Ansible. The nginx container was successfully deployed and verified, showing that Docker containers can be managed easily across nodes.